

From Parametric Curves to Parametric Surfaces

In Chapter 13 we parameterized **curves** in space with one parameter:

$$\vec{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b$$

One parameter traces out a one-dimensional object (a curve).

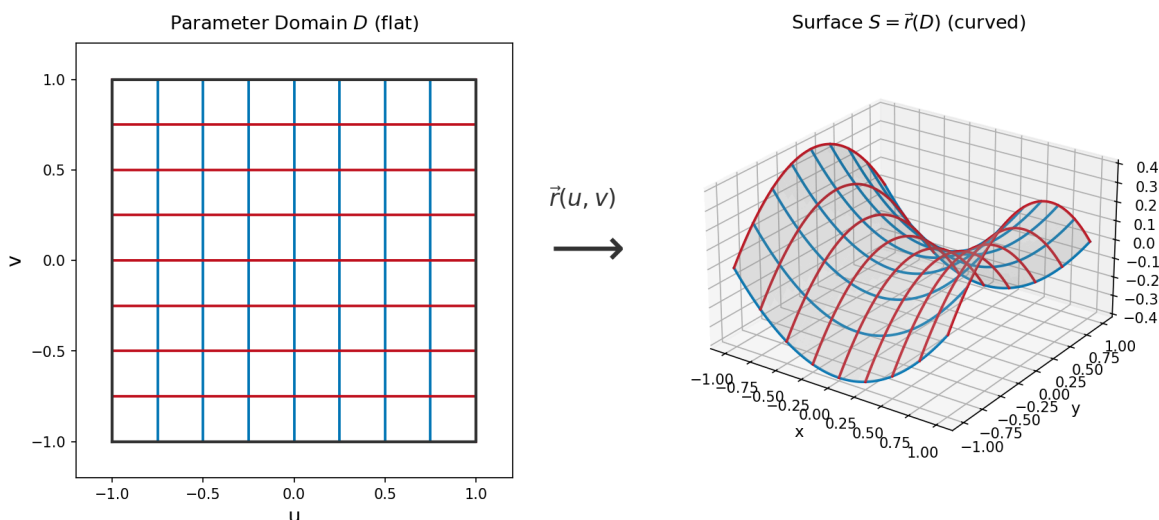
To describe a **surface** (a two-dimensional object in $\mathbb{R}^3$), we need **two parameters**:

Parametric Surface

A **parametric surface** S is defined by a vector function

$$\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$$

where (u, v) varies over a parameter domain $D \subseteq \mathbb{R}^2$.



As (u, v) ranges over D , the function $\vec{r}$ **bends the flat rectangle** into the curved surface S . The grid lines go along for the ride — the blue u -curves and red v -curves on S are the images of the grid lines in D .

One parameter $\rightarrow$ curve. Two parameters $\rightarrow$ surface. Think of $\vec{r}(u, v)$ as wrapping a flat sheet of paper (the domain D) around an object in space.

Example 1: The Unit Sphere

Example 1

Find a parametric representation for the unit sphere $x^2 + y^2 + z^2 = 1$.

Use spherical coordinates with $\rho = 1$. What are x, y, z in terms of ϕ and θ ?

Example 2: Identify the Surface

Example 2

Describe and sketch the surface with parametric equations

$$\vec{r}(u, v) = \langle u, 3 \cos v, 3 \sin v \rangle$$

For a fixed u , what does the point $(u, 3 \cos v, 3 \sin v)$ trace as v varies? What shape has $y^2 + z^2 = \text{const}$?

Quick Check: Match the Parameterization

What surface does each parameterization describe?

1. $\vec{r}(u, v) = \langle u \cos v, u \sin v, u \rangle, \quad u \geq 0, 0 \leq v \leq 2\pi$

2. $\vec{r}(u, v) = \langle u, v, 4 - u^2 - v^2 \rangle$

For (1): eliminate the parameter to find $x^2 + y^2 = ?$. For (2): what is z in terms of x and y ?

Surfaces of Revolution

Many surfaces arise by rotating a curve around an axis. The rotation angle becomes the second parameter.

Step 1: the generating curve. Take the curve $y = f(x)$ in the xy -plane, $a \leq x \leq b$.

Step 2: rotate about the x -axis. At each fixed x , the cross-section is a **circle of radius** $|f(x)|$ in the yz -plane, parameterized by angle θ :

$$y = f(x) \cos \theta, \quad z = f(x) \sin \theta$$

Step 3: assemble the full parameterization.

$$\vec{r}(x, \theta) = \langle x, f(x) \cos \theta, f(x) \sin \theta \rangle, \quad a \leq x \leq b, \quad 0 \leq \theta \leq 2\pi$$

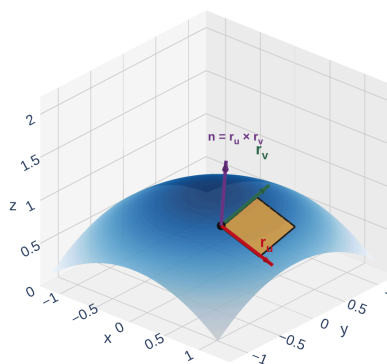
Example 3

Give parametric equations for the surface obtained by rotating $y = x(1 - x)$, $0 \leq x \leq 1$, about the x -axis.

Use the formula above with $f(x) = x(1 - x)$.

Tangent Vectors to a Parametric Surface

Tangent Vectors $\vec{r}_u$, $\vec{r}_v$ and Normal $\vec{n} = \vec{r}_u \times \vec{r}_v$



[Interactive version](#)

In Section 15.5, we found the surface area correction factor for graphs $z = f(x, y)$ by taking tangent vectors and computing their cross product. Now we generalize to **any** parametric surface.

Tangent vectors: At any point $\vec{r}(u_0, v_0)$, differentiate with respect to each parameter:

$$\vec{r}_u = \frac{\partial \vec{r}}{\partial u} = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right\rangle \quad \vec{r}_v = \frac{\partial \vec{r}}{\partial v} = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right\rangle$$

$\vec{r}_u$ is tangent to the u -curve (hold v fixed, vary u).

$\vec{r}_v$ is tangent to the v -curve (hold u fixed, vary v).

Normal vector: If $\vec{r}_u \times \vec{r}_v \neq \vec{0}$,

$$\vec{n} = \vec{r}_u \times \vec{r}_v$$

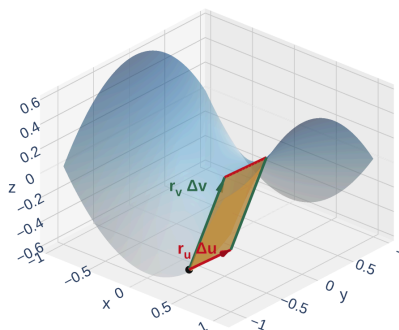
is **normal to the surface** at that point (perpendicular to both tangent directions).

$\vec{r}_u$ and $\vec{r}_v$ span the tangent plane. Their cross product $\vec{r}_u \times \vec{r}_v$ gives a normal vector. This is exactly the same idea as Section 15.5, but now for general parameterizations.

Surface Area of a Parametric Surface

Same strategy as Section 15.5: approximate each small surface patch by a parallelogram.

$$\Delta S \approx |\vec{r}_u \times \vec{r}_v| \Delta u \Delta v$$



[Interactive version](#)

Tile D with rectangles of size $\Delta u \times \Delta v$. Under $\vec{r}$, each rectangle maps to a small patch on S approximately shaped like a parallelogram.

Patch sides: $\vec{r}_u \Delta u$ and $\vec{r}_v \Delta v$. Patch area:

$$\Delta S \approx |\vec{r}_u \Delta u \times \vec{r}_v \Delta v| = |\vec{r}_u \times \vec{r}_v| \Delta u \Delta v$$

Sum and take the limit $\Delta u, \Delta v \rightarrow 0$:

Surface Area of a Parametric Surface

If S is given by $\vec{r}(u, v)$ for $(u, v) \in D$, and $\vec{r}$ is smooth, then

$$A(S) = \iint_D |\vec{r}_u \times \vec{r}_v| dA$$

Surface area = $\iint_D |\vec{r}_u \times \vec{r}_v| dA$. The cross product magnitude is the “correction factor” for how much the parameterization stretches the flat domain into the curved surface.

Quick Check: Compute the Normal Vector

For the cylinder $\vec{r}(u, v) = \langle u, 3 \cos v, 3 \sin v \rangle$, find $\vec{r}_u$, $\vec{r}_v$, and $\vec{r}_u \times \vec{r}_v$.

Differentiate each component with respect to u , then with respect to v . Then compute the cross product.

Example 4: Surface Area of a Sphere

Example 4

Use the parametric representation to find the surface area of a sphere of radius R .

Parameterization: $\vec{r}(\theta, \phi) = \langle R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi \rangle, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$

Compute $\vec{r}_\theta$, $\vec{r}_\phi$, their cross product, and its magnitude. Then integrate over the parameter domain.

Example 5: Surface Area of a Cone

Example 5

Find the surface area of the cone $z = \sqrt{x^2 + y^2}$ below $z = 3$.

Parameterize using r (radius) and θ . Write $\vec{r}(r, \theta) = \langle r \cos \theta, r \sin \theta, r \rangle$. Then find $|\vec{r}_r \times \vec{r}_\theta|$ and integrate.

Special Case: Graphs $z = f(x, y)$ Revisited

Let's verify that the parametric formula recovers our Section 15.5 result.

Parameterization of a graph: $\vec{r}(x, y) = \langle x, y, f(x, y) \rangle$

Tangent vectors:

$$\vec{r}_x = \langle 1, 0, f_x \rangle \quad \vec{r}_y = \langle 0, 1, f_y \rangle$$

Cross product:

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} = \langle -f_x, -f_y, 1 \rangle$$

$$|\vec{r}_x \times \vec{r}_y| = \sqrt{f_x^2 + f_y^2 + 1}$$

So the parametric surface area formula gives:

$$A(S) = \iint_D |\vec{r}_x \times \vec{r}_y| dA = \iint_D \sqrt{f_x^2 + f_y^2 + 1} dA$$

This is **exactly** the formula from Section 15.5!

The graph formula $\sqrt{f_x^2 + f_y^2 + 1}$ is the special case of $|\vec{r}_u \times \vec{r}_v|$ when $u = x, v = y$. The parametric formula is more general — it handles surfaces that can't be written as graphs (spheres, tori, etc.).

Quick Check: Compute the Correction Factor

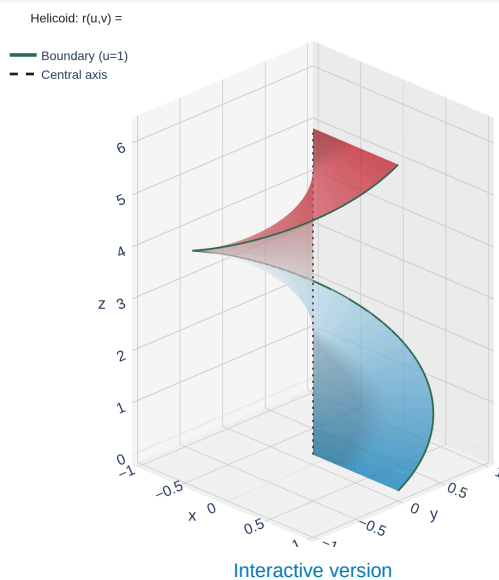
For the paraboloid $\vec{r}(r, \theta) = \langle r \cos \theta, r \sin \theta, r^2 \rangle$, $0 \leq r \leq 2$, find $|\vec{r}_r \times \vec{r}_\theta|$.

Find $\vec{r}_r$ and $\vec{r}_\theta$, compute the cross product, then take the magnitude.

Example 6: Surface Area of a Helicoid

Example 6

Find the surface area of the helicoid $\vec{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$ for $0 \leq u \leq 1, 0 \leq v \leq 2\pi$.



Compute $\vec{r}_u$, $\vec{r}_v$, then $|\vec{r}_u \times \vec{r}_v|$, and set up the double integral.

Homework

Section 16.6: 1, 3, 5, 7, 9, 13--18, 19, 21, 23, 25, 33, 35, 39, 41, 43, 45, 47, 49